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NUMERICAL SOLUTION OF SNOW MELT

GRADUATION THESIS

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SUMMARY

In this study, snow melt process considered as a one-dimensional phase change problem. The enthalpy method was used to obtain the numerical solution equations and Neumann's Solution was used for analytical solution. During numerical operations, Python and NumPy library are used.

Keywords: Heat Transfer, Phase Change, Enthalpy Method, Numerical Methods, Implicit Method

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ABBREVIATIONS

T_i^k : temperature at $t=k$ and $x=i$

k : thermal conductivity

ρ : density

c : specific heat

α : thermal diffusivity

Fou: fourier number $\left(\frac{k}{\rho * c p}\right) \frac{dt}{dx^2}$

L : Latent Heat

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INTRODUCTION

Since the melting process involves phase change, it is natural to consider heat equation for this model. As mentioned in lectures that heat equation is a form of partial differential equation, we would need the initial conditions and the boundary conditions to find a solution. Problem aims to describe the temperature distribution in a homogeneous medium undergoing a phase change. Problem of this solution is hard and complex for hand calculations. In this paper, we examine a model of snow melting through Neumann's Problem. This problem is established in Section 3, analytically solved in Section 1 and numerically solved in Section 2.

ASSUMPTIONS

- The effects of the natural convection within the melt can be ignored.
- Thermophysical properties are different for the solid and liquid phases but are independent of temperature.
- Heat transfer only occurs only on x axis

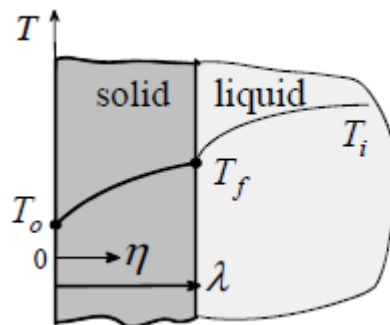
1. CONDUCTION WITH PHASE CHANGE[1]

Conduction with phase change is characterized by a moving interface. The motion and location of the interface are unknown a priori and thus must be determined as part of the solution. Since material properties change following phase transformation and since a discontinuity in temperature gradient exists at the interface, it follows that each phase must be assigned its own temperature function.

1.1 Stefan's Solution

One of the earliest published exact solutions to phase change problems is credited to Stefan who published his work in 1891. He considered solidification of a semi-infinite liquid region. The liquid is initially at the fusion temperature T_f . The surface at $x = 0$ is suddenly maintained at temperature $T_o < T_f$. Solidification begins instantaneously at $x = 0$.

1.2 Neumann's Solution: Melting of Semi-Infinite Region



Shape 1.2.1- Schematic of Neumann's Solution[2]

Neumann solved the more general problem of phase change in a semi-infinite region which is not initially at the fusion temperature T_i which is below the melting temperature T_f . The surface at $x = 0$ at a constant temperature $T_o > T_f$. A moving interface instantaneously at $x=0$ and propagates through the solid phase. Neumann's solution to this problem is based on the assumption that the temperature distribution is one-dimensional, properties of each phase are uniform and remain constant and that motion of the liquid phase is neglected. Governing equations are,

$$\frac{\partial^2 T_L}{\partial x^2} = \frac{1}{\alpha_L} \frac{\partial T_L}{\partial t} \quad 0 < x < x_i \quad (1.2.1)$$

$$\frac{\partial^2 T_S}{\partial x^2} = \frac{1}{\alpha_S} \frac{\partial T_S}{\partial t} \quad x > x_i \quad (1.2.2)$$

Boundry conditions,

- $T_L(0, t) = T_o$
- $T_L(x_i, t) = T_f$
- $T_S(x_i, t) = T_f$
- $T_S(\infty, t) = T_i$

The interface energy equation is

$$k_L \frac{\partial T_L(x_i, t)}{\partial x} - k_S \frac{\partial T_S(x_i, t)}{\partial x} = \rho_L L \frac{dx_i}{dt} \quad (1.2.3)$$

Initial conditions,

$$T_S(x, 0) = T_i$$

$$x_i(0) = 0$$

$$\eta = \frac{x}{\sqrt{4\alpha_L t}} \quad (1.2.4)$$

$$T_S = T_S(\eta) \quad (a)$$

$$T_L = T_L(\eta) \quad (b)$$

Using equation (1.2.3), (a) and (b), equations (1.2.1) and (1.2.2) transform to

$$\frac{d^2 T_L}{d\eta^2} + 2\eta \frac{dT_L}{d\eta} = 0 \quad 0 < \eta < \eta_i \quad (c)$$

$$\frac{d^2 T_S}{d\eta^2} + 2\eta \frac{\alpha_L}{\alpha_S} \frac{dT_S}{d\eta} = 0 \quad \eta > \eta_i \quad (d)$$

where

$$\eta_i = \frac{x_i}{\sqrt{4\alpha_L t}} \quad (e)$$

Solutions to c and d,

$$T_L = A \operatorname{erf}(\eta) + B \quad (f)$$

$$T_S = C \operatorname{erf}\left(\sqrt{\frac{\alpha_L}{\alpha_S}} \eta\right) + D \quad (h)$$

Applying boundry condition($T_S(x_i, t) = T_f$),

$$T_f = A \operatorname{erf} \frac{x_i}{\sqrt{4\alpha_L t}} + B$$

$$x_i = \lambda \sqrt{4\alpha_L t} \quad (i)$$

Boundry conditions in new variables,

- $T_L(0) = T_o$
- $T_L(\lambda) = T_f$
- $T_S(\lambda) = T_f$
- $T_S(\infty) = T_i$
-

$$(1.2.3) \quad k_L \frac{\partial T_L(x_i, t)}{\partial x} - k_S \frac{\partial T_S(x_i, t)}{\partial x} = \rho_L L \frac{dx_i}{dt}$$

Exact solution is,

$$T_L(x, t) = T_o + \frac{T_f - T_o}{\text{erf} \lambda} \text{erf} \frac{x}{\sqrt{4\alpha_L t}}$$

$$T_S(x, t) = T_i + \frac{T_f - T_i}{1 - \text{erf} \sqrt{\frac{\alpha_L}{\alpha_S}} \lambda} \left(1 - \text{erf} \frac{x}{\sqrt{4\alpha_L t}}\right)$$

Where λ is given by,

$$\frac{\exp(-\lambda^2)}{\text{erf} \lambda} - \sqrt{\frac{\alpha_L}{\alpha_S}} \frac{k_S}{k_L} \frac{T_f - T_i}{T_o - T_f} \frac{\exp\left(-\frac{\alpha_L}{\alpha_S} \lambda^2\right)}{1 - \text{erf} \sqrt{\frac{\alpha_L}{\alpha_S}} \lambda} = \frac{\sqrt{\pi} L \lambda}{c_{pL}(T_o - T_f)}$$

2. NUMERICAL SOLUTION WITH ENTHALPY METHOD

2.1 Finite difference approximations[3]

First order forward difference

Use a Taylor series expansion about the point x_i

$$u(x_i + \delta x) = u(x_i) + \delta x \left. \frac{\partial u}{\partial x} \right|_{x_i} + \frac{\delta x^2}{2!} \left. \frac{\partial^2 u}{\partial x^2} \right|_{x_i} + \frac{\delta x^3}{3!} \left. \frac{\partial^3 u}{\partial x^3} \right|_{x_i} + \dots$$

where δx is a small distance, and δx^2 and δx^3 are shorthand for $(\delta x)^2$ and $(\delta x)^3$.

Solve for $\left. \frac{\partial u}{\partial x} \right|_{x_i}$

$$\left. \frac{\partial u}{\partial x} \right|_{x_i} = \frac{u(x_i, \delta x) - u(x_i)}{\delta x} - \frac{\delta x}{2!} \left. \frac{\partial^2 u}{\partial x^2} \right|_{x_i} - \frac{\delta x^2}{3!} \left. \frac{\partial^3 u}{\partial x^3} \right|_{x_i} + \dots$$

In the limit as $\delta x \rightarrow 0$ the coefficients of the higher order derivative terms vanish, and

the first term on the right hand side becomes equal to the derivative on the left hand side.

Replace δx with $\Delta x = x_{i+1} - x_i$, and substitute $u_i \approx u(x_i)$

$$\left. \frac{\partial u}{\partial x} \right|_{x_i} \approx \frac{u_{i+1} - u_i}{\Delta x} - \frac{\Delta x}{2!} \left. \frac{\partial^2 u}{\partial x^2} \right|_{x_i} - \frac{(\Delta x)^2}{3!} \left. \frac{\partial^3 u}{\partial x^3} \right|_{x_i} + \dots$$

Since Δx is the only parameter under the user's control that determines the error, the truncation error is simply written

$$\frac{\Delta x}{2!} \left. \frac{\partial^2 u}{\partial x^2} \right|_{x_i} = O(\Delta x)$$

Recall that the point of Big O notation is to look at the exponent of the truncation error term.

We conclude that the error in the first order forward difference formula is linear in the mesh spacing.

$$\left. \frac{\partial u}{\partial x} \right|_{x_i} = \frac{u_{i+1} - u_i}{\Delta x} + O(\Delta x)$$

First order backward difference

Repeat the preceding analysis with a Taylor series expansion for $u(x_i - \delta x)$.

$$\left. \frac{\partial u}{\partial x} \right|_{x_i} = \frac{u_i - u_{i-1}}{\Delta x} - \frac{\Delta x}{2!} \left. \frac{\partial^2 u}{\partial x^2} \right|_{x_i} - \frac{(\Delta x)^2}{3!} \left. \frac{\partial^3 u}{\partial x^3} \right|_{x_i} + \dots$$

or

$$\left. \frac{\partial u}{\partial x} \right|_{x_i} = \frac{u_i - u_{i-1}}{\Delta x} + O(\Delta x)$$

First order central difference

Combine Taylor Series expansions for u_{i+1} and u_{i-1} to get

$$\left. \frac{\partial u}{\partial x} \right|_{x_i} = \frac{u_{i+1} - u_{i-1}}{2\Delta x} + O(\Delta x^2)$$

Second order central difference

$$\left. \frac{\partial^2 u}{\partial x^2} \right|_{x_i} = \frac{u_{i-1} - 2u_i + u_{i+1}}{\Delta x^2} + O(\Delta x^2)$$

Summary of finite difference approximations

Forward: $\left. \frac{\partial u}{\partial x} \right|_{x_i} = \frac{u_{i+1} - u_i}{\Delta x} + O(\Delta x)$

Backward: $\left. \frac{\partial u}{\partial x} \right|_{x_i} = \frac{u_i - u_{i-1}}{\Delta x} + O(\Delta x)$

Central: $\left. \frac{\partial u}{\partial x} \right|_{x_i} = \frac{u_{i+1} - u_{i-1}}{2\Delta x} + O(\Delta x^2)$

Second Order Central: $\left. \frac{\partial^2 u}{\partial x^2} \right|_{x_i} = \frac{u_{i-1} - 2u_i + u_{i+1}}{\Delta x^2} + O(\Delta x^2)$

Applying discretization to 1D heat conduction equation,

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2} \quad (2.1.1)$$

$\partial T / \partial t$ at $x = x_i$

$$\frac{\partial T}{\partial t} = \frac{T_{x_i}^{k+1} - T_{x_i}^k}{\Delta t}$$

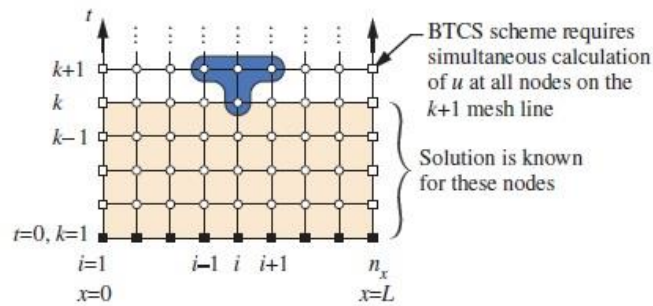
$\partial^2 T / \partial x^2$ at $t = k$

$$\frac{\partial^2 T}{\partial x^2} = \frac{T_{x_{i-1}}^k - 2T_{x_i}^k + T_{x_{i+1}}^k}{\Delta x^2}$$

Eq.(2.1.1) becomes

$$\frac{T_{x_i}^{k+1} - T_{x_i}^k}{\Delta t} = \alpha \frac{T_{x_{i-1}}^k - 2T_{x_i}^k + T_{x_{i+1}}^k}{\Delta x^2} \quad (2.1.2)$$

Implicit Discretization [4]



Shape 2.2.1[5]

Shift the k subscripts by one, and move all T^{k+1} terms to the left hand side on Eq. (2.1.2). We assume that $fou = \alpha \frac{\Delta t}{\Delta x^2}$,

$$[-fou]T_{x_{i-1}}^{k+1} + [1 + 2fou]T_{x_i}^{k+1} + [-fou]T_{x_{i+1}}^{k+1} = T_{x_i}^k \quad (2.1.3)$$

Convert to Eq.(2.2.3) to matrix form. N is number of points on domain.

$$[A]\{x\} = [B]$$

$$[A] = \left[\begin{array}{cccccc} 1 + 2f_{ou} & -f_{ou} & 0 & 0 & \dots & 0 \\ 0 & -f_{ou} & 1 + 2f_{ou} & -f_{ou} & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & -2f_{ou} & 1 + 2f_{ou} \end{array} \right] \left. \vphantom{\begin{array}{c} 1 + 2f_{ou} \\ 0 \\ \vdots \\ 0 \end{array}} \right\} N - 2$$

$$\{x\} = \begin{Bmatrix} T_1 \\ T_2 \\ \vdots \\ T_n \end{Bmatrix}$$

$$[B] = \begin{Bmatrix} T_1^k \\ T_2^k \\ \vdots \\ T_n^k \end{Bmatrix} [6]$$

2.2 Enthalpy Method[7]

The enthalpy formulation for the conduction controlled phase change can be written as,

$$\frac{\partial H}{\partial t} = \text{div} \left(\frac{k}{\rho} \text{grad} T \right) \quad (2.2.1)$$

Eq. (2.1.1) can be splitting the total enthalpy H into sensible and latent heat component.

$$H = h + L * f_l$$

Where

$$h = \int_{T_f}^T c \, dT$$

$$f_l(T) = \begin{cases} 1 & \text{if } T > T_f \\ 0 & \text{if } T < T_f \end{cases}$$

$$\frac{\partial H}{\partial t} = \frac{\partial h}{\partial t} + L \frac{\partial f_l}{\partial t}$$

Eq. (2.2.1) written as,

$$\frac{\partial h}{\partial t} = \text{div} \left(\frac{k}{\rho} \text{grad} T \right) - L \frac{\partial f_l}{\partial t} \quad (2.2.2)$$

Eq. (2.2.2) written with implicit discretization for internal nodes.

$$\frac{\partial h_i}{\partial t} = \frac{k}{\rho \Delta x^2} (T_{i-1} - 2T_i + T_{i+1}) - L \frac{\partial f_l}{\partial t} \quad (2.2.3)$$

Discretization equations for boundary nodes depend on specific boundary conditions.

Consider first the case when control volume i is fully solid or liquid. In that case,

$$\frac{\partial h_i}{\partial t} = c \frac{\partial T_i}{\partial t}$$

$$\frac{\partial f_l}{\partial t} = 0$$

Eq. (2.2.3) written as now,

$$\frac{\partial T_i}{\partial t} = \frac{k}{\rho c \Delta x^2} (T_{i-1} - 2T_i + T_{i+1})$$

Backward differencing of the left side term gives,

$$a_{i-1}T_{i-1} + a_iT_i + a_{i+1}T_{i+1} = b_i \quad (2.2.4)$$

$$a_{i-1} = a_{i+1} = -fou$$

$$a_i = 1 + 2fou$$

Now, consider the case when melting occurs around a certain node i . In that case, the liquid fraction f lies strictly in the interval $[0,1]$.

$$T_i = T_f$$

$$\frac{\partial h_i}{\partial t} = 0$$

$$L \frac{\partial f_{li}}{\partial t} = \frac{k}{\rho \Delta x^2} (T_{i-1} - 2T_i + T_{i+1})$$

Backward differencing of the liquid fraction term gives:

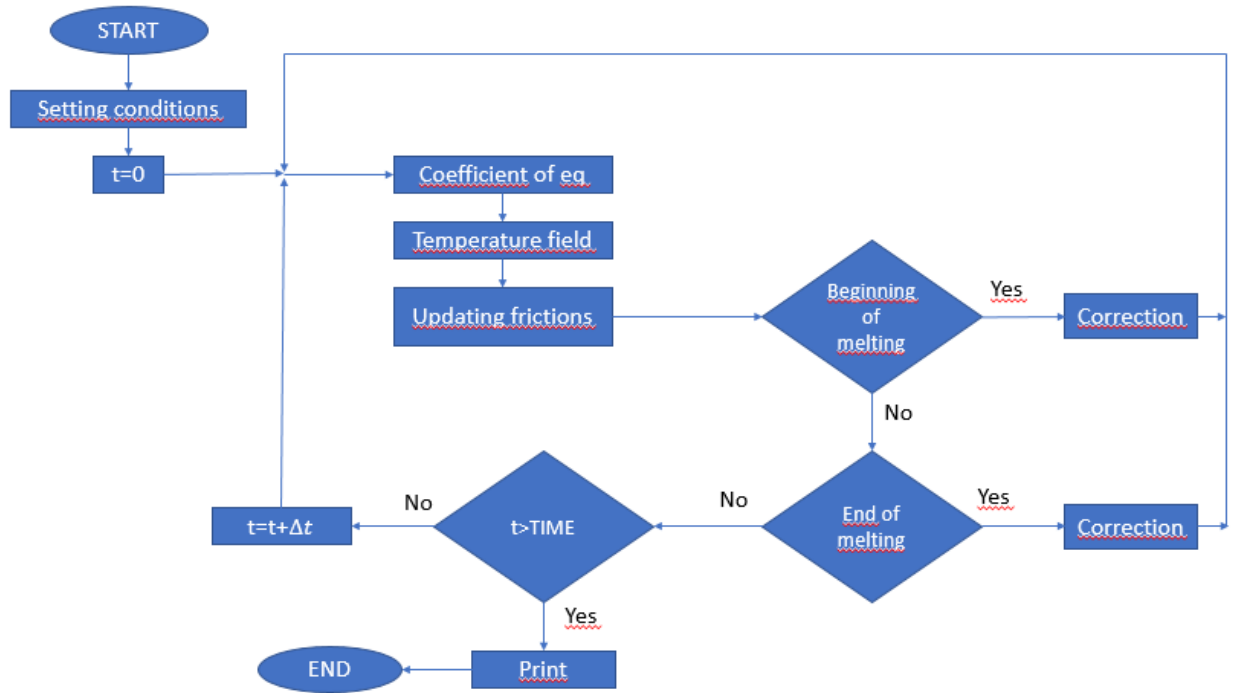
$$f l_i = f l_{i_{old}} + \frac{k}{L \rho c \Delta x^2} (T_{i-1} - 2T_m + T_{i+1}) \quad (2.2.5)$$

Liquid fraction terms are updated from temperature field and not from the sensible enthalpy field. The temperature field can be calculated by the Eq. (2.2.5) using,

$$a_{i-1} = a_{i+1} = 0$$

$$a_i = 1$$

Flow-chart for conduction controlled phase change,

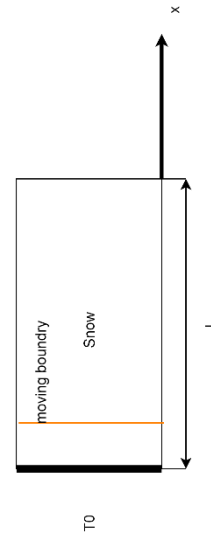


Shape 2.2.2

3. SNOW MELTING

3.1 Problem definition

$$L=2m$$



Shape 3.1.1

Thermophysical properties of snow and water

Material	$k(\text{W/mK})$	$\rho(\text{kg/m}^3)$	$c_p \left(\frac{\text{kJ}}{\text{kgC}} \right)$
Snow	0.3	400	2.090
Water	0.6	997	4.184

Table 3.1.1[8]

Initial conditions,

$$T(x, 0) = -10^\circ\text{C}$$

$$x_i(0) = 0 \text{ m}$$

Boundry conditions,

$$T(0, t) = 20^\circ\text{C}$$

$$T(x_i, t) = T_f$$

$$T(\infty, t) = -10^\circ\text{C}$$

$$T_f = 0^\circ\text{C}$$

3.2 Solution with codes

Analytical Solution in MATLAB

syms λ

```

cps=2.090; %specific heat of solid phase
cpl=4.184; %specific heat of liquid phase
ks=0.3;    %thermal conductivity of solid phase
kl=0.6;    %thermal conductivity of liquid phase
Tf=0;      %fusion temperature
To=20;     %boundry temperature
Ti=-10;    %initial temperature
rhoL=997;  %density of liquid phase

```

```

rhos=400; %density of solid phase
al=kl/(rho_l*cpl); %thermal diffusivity of liquid phase
as=ks/(rhos*cps); %thermal diffusivity of solid phase

L=334; %Latent heat
t=(4.32)*50; %Time

%Neumann's Solution
flambda=exp(-lambda^2)/erf(lambda)-(sqrt(al/as))*(ks/kl)*((Tf-Ti)/(To-
Tf))*(exp(-lambda^2*al/as))/(1-erf(sqrt(al/as)*lambda))...
==(sqrt(pi)*L*lambda)/(cpl*(To-Tf));

symlambda=vpasolve(flambda,lambda); %Equation Solver
dblambd=double(symlambda); %Symbolic variable to double variable
xi=dblambd*sqrt(4*al*t) %moving boundry index in domain
x=0:0.001:2; %nodes
Tl=To+((Tf-To)/erf(dblambd))*erf(x/sqrt(4*al*t)); %temperature
distribution of liquid phase
Ts=Ti+((Tf-Ti)/(1-erf((al/as)*dblambd)))*(1-erf(x/sqrt(4*as*t)));
%temperature distribution of solid phase
layerindex=uint16(xi/0.001); %moving boundry index in arrays

Tlp=Tl(1:layerindex); %seperating temperature distrubition of
liquid phase for plotting
Tsp=Ts(layerindex:end); %seperating temperature distrubition of
solid phase for plotting
plot(x(1:layerindex),Tlp,x(layerindex:end),Tsp) %plotting temperature
distrubition of problem
legend('Liquid Phase','Solid Phase') %plotting legends

```

Numerical solution with Python

```

import numpy as np
import matplotlib.pyplot as plt
import scipy.io

#problem boundaries

dx=0.1 #dx
Long=dx*(2/dx) #domain length
dt=40 #dt
time=dt*50

#thermophysical properties
#rho=8920
k_s=0.3 #thermal conductivity of solid phase
k_l=0.6 #thermal conductivity of liquid phase
#cp=583.45
Tf=0 #fusion temperature
L=334 #latent heat
rho_s=400 #density of solid phase
rho_l=997 #density of liquid phase
cp_l=4.184 #specific heat of liquid phase

```

```

cp_s=2.090    #specific heat of solid phase

#variables
nx=int(Long/dx)+1    #number of points
nt=int(time/dt)      #time step
alpha_s=k_s/(rho_s*cp_s) #thermal diffusivity of solid phase
alpha_l=k_l/(rho_l*cp_l) #thermal diffusivity of liquid phase
Fou_s=alpha_s*dt/dx**2    #fou number of solid phase
Fou_l=alpha_l*dt/dx**2    #fou number of liquid phase
z=(k_l*dt)/(rho_l*L*dx**2) #liquid fraction coefficient

#Implicit Method for Calculating Temperature Distrubition
def temp_field(a_1,a_2,a_3,Told,nx):
    #Coefficients
    A=np.full((nx-1,nx-1),0.0) #Coefficient Matrix
    r=0
    for c in range(nx-1):
        if c==0 and r==0:
            A[r][c]=a_2
            A[r][c+1]=a_3
            r=r+1
        elif c==nx-2 and r==nx-2:
            A[r][c]=a_2
            A[r][c-1]=2*a_1
            r=r+1
        else:
            A[r][c-1]=a_1
            A[r][c]=a_2
            A[r][c+1]=a_3
            r=r+1
    t_val=np.linalg.solve(A,T[1:])
    return t_val

t=0 #time starts from 0
Ti=-10 #initial temperature
init_temp=Ti*np.ones(nx) #initial temperature array
T=init_temp.copy() # creating temperature array(T)
T[0]=20 # boundry condition of problem
melting_temps=Tf*np.ones(nx) #Melting temperature array for melting process
fl=np.zeros(nx) #liquid fraction term
a=np.array([1,1,1]) #coefficient array

while(t<=time):
    Told=T.copy()
    fl_old=fl.copy()
    for i in range(1,nx-1):

        if T[i]<Tf:

```

```

a[0]=-Fou_s
a[1]=1+2*Fou_s
a[2]=a[0]
b=Told.copy()
T[1:]=temp_field(a[0],a[1],a[2],b,nx)
t=t+dt
if int(T[i])==Tf:
    a[0]=0
    a[1]=1
    a[2]=a[0]
    b=melting_temps.copy()
    T[1:]=temp_field(a[0],a[1],a[2],b,nx)
    fl[i]=fl_old[i]+z*(T[i-1]-2*Tf+T[i+1])
    t=t+dt

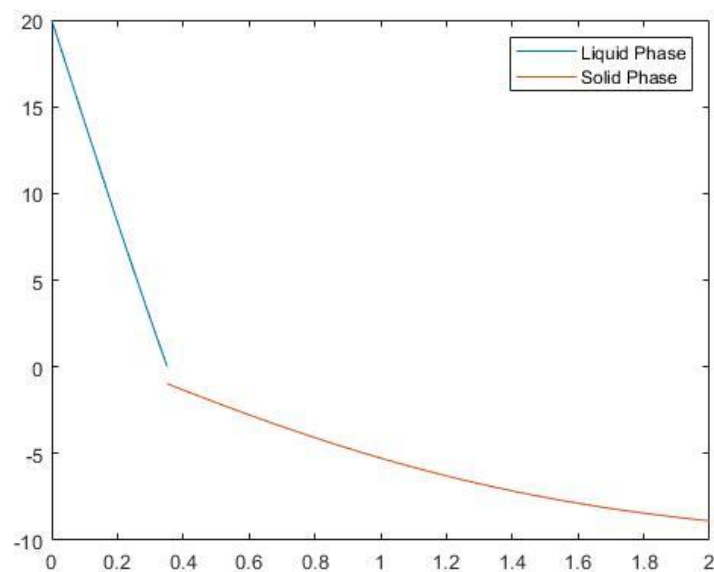
x_values=np.linspace(0,Long,nx)    #x axis elements
plt.xlabel("x values")
plt.ylabel("Temperature(°C)")
plt.ylim([-11,21])
plt.xticks(np.arange(0, Long, 0.2))

plt.plot(x_values,T,)    #plotting value

```

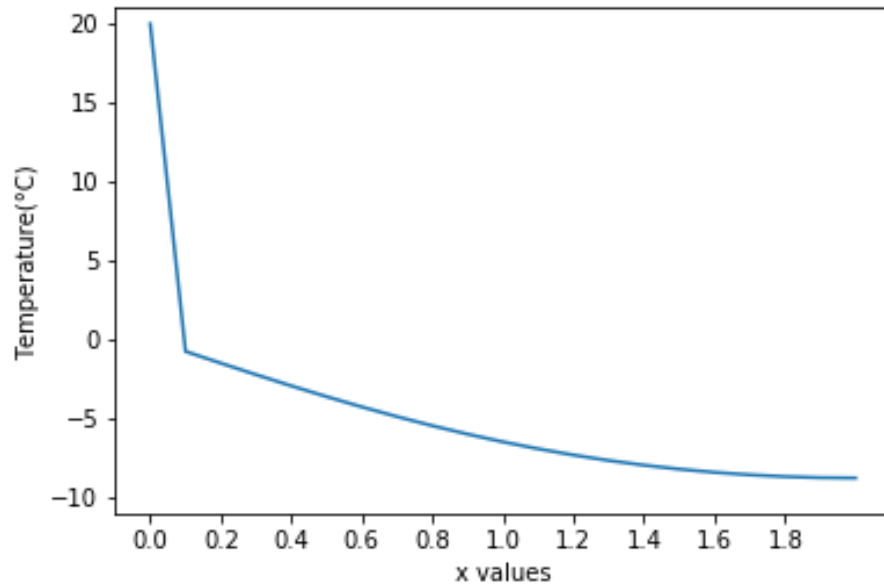
4. RESULTS AND CONCLUSION

According the analytical solution in MATLAB moving boundry layer is in $x= 0.3516$ and temperature distrubition of problem as below ,



Schape 4.1

The code written for the numerical solution did not work as it should. The liquid fraction values remained constant when they should have changed. A difference was observed between the analytical solution and the numerical solution in the temperature distribution. These results show that the usage and implemented method for Python does not work correctly.



Shape 4.2

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